

Arithmetic Sequences and Series

ANSWER KEY



Arithmetic sequences

- 1 Consider the sequence 7, 11, 15, 19, ...
- a** State the common difference. $d = 4$
- b** Write a formula for the n th term. $a_n = 4n + 3$
- c** Find a_{20} . $a_{20} = 83$
- 2 In an arithmetic sequence, $a_3 = 10$ and $a_8 = 25$.
- a** Find the common difference and the first term. $5d = 15$ gives $d = 3$; $a_1 = 10 - 2(3) = 4$.
- b** Write a formula for a_n . $a_n = 4 + 3(n - 1) = 3n + 1$.
-

Arithmetic series

- 3 Find the sum of the first 30 terms of the series $5 + 9 + 13 + \dots$
- $a_{30} = 5 + 29(4) = 121$, so $S_{30} = \frac{30(5 + 121)}{2} = 1890$.
- 4 Consider the multiples of 6 strictly between 100 and 500.
- a** How many are there? First 102, last 498: $n = \frac{498 - 102}{6} + 1 = 67$.
- b** Find their sum. $S = \frac{67(102 + 498)}{2} = 67 \times 300 = 20,100$.
- 5 A theatre has 18 seats in the first row, and each subsequent row has 3 more seats than the row before. How many seats are there in total in 25 rows?
- $a_{25} = 18 + 24(3) = 90$, so $S_{25} = \frac{25(18 + 90)}{2} = 1350$ seats.
-

Arithmetic means and sequence properties

- 6 Insert three arithmetic means between 5 and 25 (i.e., find b_1, b_2, b_3 so that 5, $b_1, b_2, b_3, 25$ is arithmetic). 5 terms total, so $d = \frac{25 - 5}{4} = 5$. Means: 10, 15, 20.
- 7 Given the arithmetic sequence with $a_1 = -8$ and $d = 3$, find the smallest n such that $a_n > 100$.
- $a_n = -8 + 3(n - 1) = 3n - 11 > 100$: $3n > 111$, $n > 37$: smallest integer $n = 38$.
- 8 Prove that the sum of the first n positive odd integers ($1 + 3 + 5 + \dots$) equals n^2 , using the arithmetic series formula.
- The n th odd number is $a_n = 2n - 1$, with $a_1 = 1$, $d = 2$. $S_n = \frac{n(a_1 + a_n)}{2} = \frac{n(1 + 2n - 1)}{2} = \frac{n(2n)}{2} = n^2$.
-

Applications

- 9 A person saves 100 riyals in week 1, and increases savings by 20 riyals each subsequent week. Find the total saved after 20 weeks.

$$a_{20} = 100 + 19(20) = 480. S_{20} = \frac{20(100 + 480)}{2} = 5800 \text{ riyals.}$$

- 10 A stack of logs has 20 logs in the bottom row, 19 in the next, decreasing by 1 each row, ending with 1 log in the top row. Find the total number of logs.

$$\text{This is the sum } 1 + 2 + \cdots + 20 = \frac{20(1 + 20)}{2} = 210 \text{ logs.}$$

Visualizing an arithmetic sequence

- 11 The first 6 terms of the sequence $a_n = 3n - 1$ are plotted below. Use the graph to confirm the sequence is linear in n , and state the common difference from the graph.

The points $(1, 2), (2, 5), (3, 8), (4, 11), (5, 14), (6, 17)$ lie on a straight line, confirming the linear (arithmetic) pattern. The common difference $d = 3$ matches the constant rise between consecutive points.

Arithmetic sequences in geometry

- 12 The interior angles of a pentagon form an arithmetic sequence. If the smallest angle is 80° and the angles sum to 540° (the pentagon angle sum), find the common difference.

$$S_5 = \frac{5(2a_1 + 4d)}{2} = 540:$$
$$\frac{5(160 + 4d)}{2} = 540 \Rightarrow 5(160 + 4d) = 1080 \Rightarrow 160 + 4d = 216 \Rightarrow 4d = 56 \Rightarrow d = 14^\circ.$$

Finding a term given the sum

- 13 The sum of the first n terms of an arithmetic sequence with $a_1 = 3$ and $d = 4$ is $S_n = 210$. Find n .

$$S_n = \frac{n(2(3) + (n-1)4)}{2} = \frac{n(4n+2)}{2} = n(2n+1) = 210 \Rightarrow 2n^2 + n - 210 = 0. \text{ Quadratic formula:}$$
$$n = \frac{-1 \pm \sqrt{1 + 1680}}{4} = \frac{-1 \pm 41}{4}. \text{ Taking the positive root: } n = 10.$$

Arithmetic sequences with negative common difference

- 14 A sequence begins 50, 44, 38, 32, ... Find the first negative term.

$$d = -6: a_n = 50 - 6(n-1) = 56 - 6n. \text{ Need } 56 - 6n < 0 \Rightarrow n > 9.33: \text{ first negative term is}$$
$$a_{10} = 56 - 60 = -4.$$

Proving an arithmetic sequence property

- 15** Prove that for any three consecutive terms a_{n-1} , a_n , a_{n+1} of an arithmetic sequence, the middle term equals the average of its neighbors: $a_n = \frac{a_{n-1} + a_{n+1}}{2}$.
- By definition, $a_{n-1} = a_n - d$ and $a_{n+1} = a_n + d$. So $\frac{a_{n-1} + a_{n+1}}{2} = \frac{(a_n - d) + (a_n + d)}{2} = \frac{2a_n}{2} = a_n$, confirming the middle term is the average of its neighbors.